Momentum Trading Strategy using Machine Learning – Hackathon Problem Statement

Objective

Build and backtest a machine learning-based long-only momentum strategy in Python using daily stock data.

Universe

Use the following 10 stocks(download daily data from Yfinance):

AAPL, MSFT, GOOGL, AMZN, META, TSLA, JPM, V, JNJ, BRK.B

Data

Daily OHLCV data

Time Period: 2017–2025

Challenge

Predict whether each stock will generate a positive return over the next week.

1. Rank all 10 stocks each week by predicted probability of positive next-week return.
2. Select the top 2 stocks.
3. Construct an equal-weight long-only portfolio of these 2 stocks.
4. Hold the portfolio for 1 week and rebalance at the start of the next week.
5. Include transaction costs of 0.1% at entry and 0.1% at exit.

Modeling

Build at least one classification model (e.g., Logistic Regression, Random Forest, XGBoost). Ensemble or voting classifiers are optional.

Validation

Recommended setup:

Train on 2017–2022

Test on 2023–2025

Advanced teams may implement walk-forward retraining.

Deliverables

1. Jupyter Notebook

- data cleaning and feature engineering
- model training and predictions
- portfolio construction and backtest
- performance analysis and visualizations

2. Short Report / Slides

- model used
- key features
- validation setup
- major findings

3. CSV File

- weekly stock predictions
- selected stocks and portfolio weights

Metrics

Report the following:

- cumulative return
- annualized return

- annualized volatility

- Sharpe ratio

- max drawdown

- hit rate

- average weekly return

Show results before and after transaction costs.

Evaluation Criteria

Criteria	Weight
Modeling approach and feature engineering	30%
Portfolio performance	30%
Backtesting correctness	20%
Clarity of analysis and presentation	10%
Code quality and reproducibility	10%